

ULTIMATE REALITY

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ABSTRACT

This paper establishes the individual's immediate reality of own presence as the foundational and fundamentally irrefutable premise of knowledge, operating through the self-sufficient mutual definition of identity and difference. The author argues that any ontological framework is inherently rooted in this reflexive architecture. Analysis of higher-order reflections demonstrates how the structure of spacetime emerges as a fractal manifold marked out on a continuous projective space. Finally, the concepts of teleology (the ultimate goal) and the post-mortem personality are deduced as logical extensions of this ontological system.

THE PREMISED POINT: "I"

I know that I exist, and that's all I know for sure.
Otherwise, who doubts or claims the opposite?
This knowledge represents me as my reflection.

Reflection identifies and distinguishes me from myself.
I am the one who sees myself as a different one.

Everything appears to me through myself, and may be merely my own self-discrimination.
So, reflection is also the foundation of any knowledge, and this is also the sole sufficient foundation.

BEING IS CONSCIOUSNESS

There is no being without properties, and properties are revealed through relationships. That which is not related to anything does not exist. Existence is a relationship between sides.

The sides are distinct, otherwise there would be nothing to relate to, and they coincide, otherwise there would be no contact between them. Therefore, a relationship is an identity of difference and a difference of identity, just like reflection. And therefore, any existence is a reflection of the sides of a relationship.

It is accessible to me only through my reflection, and I cannot separate it from my own reflection, since distinction is impossible without identification, and any distinction can come from the identity of my reflection.

Reflection is a knowledge one with oneself (in Russian literally consciousness). Consequently, consciousness is simultaneously existence as being and as a knowledge of being (in Russian literally co-being, that is event).

And this is the only being and knowledge: one can know only what exists, and exists only what can be known . Even the absolute thing-in-itself exists through self-knowledge—its own reflection.

DIALECTICS

Identity defines itself through difference: only that which is not distinct can be distinguished. Difference defines itself through identity: only that which is not identical can be identified. Identity and difference are defined mutually, in each other, and require nothing else for their definition.

They do not exist separately from each other, but together they form a self-determined and thus self-sufficient and self-existent consciousness-being.

It is not conditioned by anything else, since there is no “other.” If there were non-being, it would by definition be being, and if it did not exist, it would be the non-being of non-being, that is, being.

Therefore, **I = reflection = relation = is the only reliable, unconditioned and self-sufficient existence**

It has been established that **relationship is identity of difference and difference of identity**

Other dialectical pairs are defined in the same way: **one and many, constancy and change, actuality and potentiality. Each of them represents a relationship**, and each of them is redefined through the totality of all the others.

The formal logic of the excluded middle is inapplicable to all of them. Their logic is the logic of reality “both-and”, not the “either-or” logic on which all science and practice are based.

TIME IS SELF-AWARENESS

Consciousness exists in relation to itself

In this self-relation each of two sides of the relation enters into a relationship with two “others.” Therefore, each side must double.

Since it does not exist in itself, its doubling signifies the non-existence of non-existence. That is, it becomes autonomously existing — it becomes a relation.

In the second order reflection — self-awareness — four sides of relation participate, and thus they create four new relationships.

Each of them exists independently, on its own, and also in relationship with two other sides of the reflective relationship. And these other two exist in relation to it, but not for each other.

And such a scheme precisely describes how **past, present and future** behave.

As a result of self-awareness, there appear four present moments, each with its own past and future, of which every two exist for each other not directly, but only through the past and future.

ACTUAL INFINITY

The reciprocal categories of one and many mathematically represent an actually infinite set

The four present moments of time are relations, and as such are elements of this set. Thereby, each of them contains the entire set, and is itself contained within each of them.

At the same time, any pair of them exists for each other only potentially, and is actualized in the multitude through this potentiality. The actualization of the present through potentiality realizes the “movement of time.”

In its actual aspect, the present moment is outside of this movement and can be interpreted as **space**. This space, therefore, does not exist in its own being, but in mutual realization with the passage of time, as automorphism, designated as spatial displacement.

Each of the four moments, in turn, reflects itself and creates four new ones, and so on ad infinitum, which actually exists through this potential unfolding.

Potentiality actualizes the entire actual infinity at once, in which every moment is simultaneously the same and different. Nevertheless, the “four-dimensional” structure of each act of self-consciousness must underlie the structure of space-time and its physical expression in the form of quantum fields and particles.

MATH INTERPRETATION

Numerical description

If we expand the concept of number, suitable tools emerge:

- **Non-standard numbers.** In non-standard analysis, “monads” are infinitesimal neighborhoods that contain micro-micro-monads extending into infinite depth. This is similar to moments, each of which contains an entire world with the same structure. However, monads exist over a continuous field of real numbers, whereas here the structure is discrete and holistic.
- **Hypercomplex numbers** (e.g., quaternions, octonions). Their algebra has several “imaginary units” that are cyclically connected and do not commute, resembling a closed quadruple of moments, each connected to the other two. However, they do not possess an internal reflexive hierarchy.
- **Ordinals and non-well-founded sets as “numbers.”** In anti-foundation theory, hypersets are defined by equations as $X = \{X, Y, Z, \dots\}$. Such objects behave almost like

symbols with self-membership. If each moment is assigned a similar hyperset, the relations between them are defined by a system of equations. This is almost a “numerical” description, but the numbers here are not quantitative measures, but graphs with cycles. They can be encoded by adjacency matrices, that is, ultimately, by ordinary numbers (0 and 1). However, this description is static; it does not convey the “becoming” of the relation. [Aczel, P. (1988)]

Thus, a numerical description is possible as a symbolic encoding of the structure of connections, but it loses the dialectic of reflection and formation if special transition operators are not introduced.

Fractal manifolds and surfaces of infinite genus.

- Imagine a surface that, at each point, locally looks like four converging petals, with each petal, when magnified, proving to be the same surface. This is the **geometric analogue of a holonic quadruple**: the whole consists of parts, each of which is a whole. An example is the generalized Sierpinski carpet or the limit sets of Kleinian groups that generate self-similar patterns. They are topologically continuous but have fractional dimension.
- **Solenoids and foliations with Cantor transversals.** A solenoid is a space that looks like an infinite spiral winding around a torus, with each turn containing infinitely many smaller turns. If we consider a solenoid over a quadruple of circles glued together in a special way, we obtain a model in which each local piece is structured as four intertwined times, while globally everything is continuous and contains an infinite number of self-similar “copies.” Solenoids are both actually infinite-dimensional and compact, which is ideal for describing an actually infinite set that exists “at once.”
- **Manifolds with self-membership (non-commutative geometry).** In classical geometry, a point either belongs to a surface or it doesn’t. But here, “every moment contains the entire set.” To convey this, we need a space where the concept of a point is blurred and replaced by a “quantum” neighborhood capable of containing the whole. This leads to non-commutative spaces [Connes, A. (1994)], where coordinates become operators, and “points” lose their individuality. Such a geometry is no longer a surface in the usual sense, but remains formally continuous (has a spectrum, measure, and metric). It can model a situation in which four “regions” intersect pairwise such that each contains the other two, yet together they form an indecomposable fabric without individual elements.

- **Dynamical systems with attractors.** Strange attractors (e.g., the Lorenz attractor) are continuous sets consisting of an infinite number of inseparable trajectories that constantly shuffle and return. If we associate “moments” with the trajectories, their mutual interweaving can model the connectivity of quadruples through common generating elements. However, such a model describes chaotic motion rather than a cyclic structure with distinct four positions.

So, a continuous surface as a classical smooth manifold is unsuitable because it does not admit the relationship “a part is equal to and contains the whole.” But if we take a fractal set, a solenoid, or switch to non-commutative geometry, we obtain a suitable language. Such a “surface” will no longer be a two-dimensional sheet in space, but a topological structure in which the local and the global are identified, and movement along it corresponds to transitions between the past and future of different moments.

Configuration in projective space

In a quadruple of moments, each is related to the other two, while two remain unrelated. In the projective plane, this is a quadrilateral with the diagonals removed. But classical axiomatics (a line passes through any two points) restores the missing connections, making all points direct neighbors.

But in the three-dimensional projective space RP^3 does an ideal model emerge: the tetrahedron as a configuration of lines. Imagine moments not as points, but as edges of the tetrahedron, forming a closed loop of four edges (e.g., AB, BC, CD, DA). Each edge has two adjacent lines intersecting it at its vertices (the past and future), and one opposite edge that crosses it — that is, does not intersect and does not lie on the same plane. They literally “do not exist for each other” directly, but are connected through shared vertices with their neighbors. Thus, the tetrahedron provides a pure geometric realization of a four-part time loop with intersecting “non-relations.” [Coxeter, H. S. M. (1998)]

Generalizations can preserve projective clarity but introduce self-belonging:

- The infinite-dimensional projective space RP^∞ .
It has the property of self-similarity: RP^∞ contains copies of itself as subspaces, and its homotopy groups are cyclic. It is possible to construct a process in it where a point is revealed as RP^∞ , but this is still not an identity, but an embedding.

- Projective spaces over hyperfields or in noncommutative geometry, where the concept of a point is replaced by an operator. It is here that the boundary between a point and a space is erased, and one can obtain a “projective space” that is an element of itself, realized as a projective module over an operator algebra.

The infinite-dimensional projective space RP^∞ itself turns out to be a fractal in the topological and structural sense. If we take a subspace within RP^∞ , it will again be RP^∞ (or at least homeomorphic to it).

RP^∞ is a smooth fractal: it has no fractional dimension in the usual sense (its dimension is infinite and integer, like that of a manifold), but it is self-similar, contains a reduced version of itself, and can be “unfolded” infinitely, revealing ever newer layers isomorphic to the whole. This corresponds precisely to the fact that a moment, being RP^∞ , reveals within itself an identical RP^∞ populated by moments.

- Projective space defines the rule of incidence—how elements are nested and dual. It is a continuous background.
- Fractal structure emerges from the act of reflexive generation — each moment, negating itself (doubling), transforms into two relations, which again transform into moments, and so on. This process generates a discrete network of moments, which, however, is marked out on a continuous projective manifold and completely replicates its topology of self-similarity. Fractal Cantor dust arises if we record only discrete “acts of self-awareness,” cutting out individual points-moments from the continuous projective fabric. However, they are not completely separated, but form a continuous whole — thus, what we get is not dust, but precisely RP^∞ .

Indra’s Mirror

In a typical fractal (Cantor dust, Sierpinski triangle), self-similarity is a function from the whole to its proper subsets. The arrow always points from the larger to the smaller. There is no closure.

The projective space RP^∞ is self-similar in the sense that it contains subspaces isomorphic to the entire RP^∞ . But here, too, the embedding is one-sided: a subspace is a part, not an element. An element of RP^∞ is a point (a line in infinite-dimensional space). The point itself does not unfold back into the entire RP^∞ . For it to unfold, each point

must simultaneously be the entire space. This requires rejection the fundamental distinction between “element and set.”

A non-founded fibration, or hyperset with a projective structure. Each element (moment) is a hyperset, which at one level is a point, and at another, the entire space. Its intrinsic geometry is a projective space, but incidence is replaced by a membership relation with loops. A point belongs to a line not as an element of a set, but as part of a self-belonging structure.

- It is infinite-dimensional and self-similar (like RP^∞).
- It contains closed membership loops: the whole is embedded in the part, and the part is embedded in the whole.

These two properties do not contradict each other because the very concept of a “point” as a separate entity disappears. The hyperset itself, unfolded into projective space, becomes a point, and simultaneously, the point unfolds into this same space.

Algebra of the Moment

Quaternions. They are cyclically closed: $ij = k$, $jk = i$, $ki = j$, and yet anticommute ($ji = -k$). This precisely describes the tripartite rhythm of time within a single “now”:

- One unit defines the “present $\leftrightarrow$ past” axis.
- Another defines the “present $\leftrightarrow$ future” axis.
- The third arises as their composition and represents the “crossed” relationship that is neither directly past nor future, but connects their cycle.

Spinors. Rotation in three-dimensional space (like quaternions acting on vectors) is a single reflection of a relation. But a spinor is an object that changes sign when rotated 360 degrees, and returns to itself when rotated 720 degrees. This is the formal expression of the “non-existence of non-existence”: the first negation reverses, the second restores, but on a new level, as autonomous existence.

A moment is not a point, but a double covering of itself. Quaternions provide the algebra of this rotation, and spinors provide what rotates. Within a single moment, a spinor already exists, distinguishing between the two sides (past/future) as two projections of a single internal rotation.

Projective space RP^3 . The group of unit quaternions is the three-dimensional sphere S^3 . The projective space RP^3 is obtained from S^3 by identifying opposite points. This is precisely the space in which spinor duality “collapses” into visible geometry: two spinor rotations (720°) yield one rotation in the projective world.

Thus, a single moment reveals itself as S^3 — a world of pure spinor rotations, where everything is doubled and reflexively closed — and is then projected into RP^3 — a world of distinguishable “present moments” and their geometric relations (crossings, incidences).

Limitations and the need for expansion. Quaternions and spinors describe a single moment without self-membership. To introduce a membership loop, we need to switch to hyperquaternions, which are multiplied not only by each other but also by themselves as elements of the enclosing set.

In non-commutative Connes geometry, spinors become operators on a Hilbert space, and a “point” can then be both an element of the algebra and a state on it. Such a spinor operator can contain the entire world as its internal state — which precisely satisfies the original requirement. [Connes, A. (1994)]

A quaternion and a spinor transform a moment from a point into a rotating monad,

Review

- The original description — a reflexive quaternary with self-membership — recognized itself as an non-well-founded hyperset with cyclic geometry.
- Projective space turned out to be its natural geometric body, and RP^3 a precise model of the crossing and duality relations within the quaternary.
- The fractal provided a language for infinite self-similarity, but did not capture the closed loop of “whole as part of part,” which required going beyond classical fractals into a hyperset monad.
- Quaternions and spinors finally gave an algebraic pulse to the individual moment: rotation, doubling, the non-existence of non-existence, the distinction of time axes — all of this turned out to be encoded in their structure as in genetic code.

Formally, it would look like this:

- **Base object:** Non-well-founded hyperset H

- **Local moment:** Coalgebraic knot m_i
- **Relations:** Noncommutative operators R_{ij}
- **Geometry:** Projective perspective space RP^3
- **Internal symmetry:** Spinor cover $SU(2)$
- **Time dynamics:** Partial order of actualizations
 $m_i \prec m_j (m_i \prec m_j)$

This is formalizable not as a single theory, but as a hybrid:

- hyperset theory
- coalgebra
- projective geometry
- spin geometry
- noncommutative algebra
- temporal logic.

These areas deal with structures where the relationship is primary, not the object, and identity arises through transformation.

FROM DIALECTICS TO ALGEBRA

Let's write identity and difference as $I = D(D)$; $D = I(I)$, and their composition into an actually infinite set as 1 (the entire set in a single element) and 0 (the "empty set" of disparate elements, infinity "compressed" into zero).

For operations with them to become an adequate expression of their dialectic, we must accept:

- $0 = 1$.
- Rejection of the axioms of Boolean algebra. Idempotency ($I \cup I = I$), distributivity, and the law of excluded middle are not required. I and D are understood as reflexive operators, obeying only given relations.
- Binary operations $+$ and $\cdot$ (not necessarily addition and multiplication of numbers, but associative, commutative, distributive abstract operations), for which $I \cdot D = 1$ and $(I + D) = 0$.

- Hence, I and D are the roots of the equation $x^2 + 1 = 0$ (if the operations behave as in a field). Then $I = i$, $D = -i$ — as a consequence.

$(I + D) = 0$ means $I \cup D = \emptyset$, but since $\emptyset = \Omega$, this also means $I \cup D = \Omega$ (completeness).

$I D = 1$ means $I \cap D = \Omega$, which for $\Omega = \emptyset$ gives $I \cap D = \emptyset$.

The two expressions do not contradict each other, but express the dual nature of reflection: it simultaneously unites and distinguishes I and D, and in the limit of actual infinity, these actions become identical. The algebraic notation $I + D = 0$ and $ID = 1$ proves to be a compact symbol of this dialectical unity.

This approach

- does not introduce unnecessary entities (complex numbers are not postulated, but emerge).
- is in keeping with the spirit of dialectics: from the relations I and D (via + and $\cdot$), their imaginary nature is deduced—this is a result, not a premise.
- allows I and D to be interpreted as reflexive operators, not as numbers.
- leaves open the possibility of generalization (for example, to quaternions, where $i^2 = j^2 = k^2 = -1$, but $ij = k \neq ji$).

Another way: special mapping of logical operations into algebraic ones.

- We define a mapping φ from the set $\{\emptyset, I, D, \Omega\}$ (emptiness, identity, difference, universe) to complex numbers: $\varphi(\emptyset)=0$, $\varphi(\Omega)=1$, $\varphi(I)=i$, $\varphi(D)=-i$.
- We interpret the union $I \cup D$ as $\varphi(I) + \varphi(D)$ (and not as the image of the union), and the intersection $I \cap D$ as $\varphi(I) \varphi(D)$.
- $\varphi(I \cup D)=i + (-i)=0 = \varphi(\emptyset)$ and $\varphi(I \cap D)=i \cdot (-i) = 1 = \varphi(\Omega)$.
- Then $I + D = i - i = 0$ and $I D = i (-i) = 1$ automatically.

Both methods are symmetrical in the sense that they describe the same algebraic structure (the field of complex numbers). Any statement derived in one method can be translated into the other. However, symmetry does not imply identity of the methods of definition: the first method defines the structure implicitly (axiomatically), the second explicitly (by model). In mathematics, both are equally valid, but in the formalization of a philosophical text, where the derivation of imaginary values from simpler principles is important, the first method is preferable.

This formalization works only within the framework of a given ontology (actual infinity, “both-and” logic, the coincidence of 0 and 1). It does not imply conformity with ordinary mathematics, only consistency with a given extraordinary ontology.

Quaternions

Let I_1, I_2, I_3 be different types of “counter-actions” (i, j, k). For each of them, the condition $I^2 = -1$ is met. But their interaction generates not 1 or 0, but each other.

In complex numbers $ID = DI = 1$. If I is a rotation by 90° in one plane, and D in another, then ID will differ from DI . This is the birth of a quaternionic structure.

The “invariant of actual infinity” $ID = 1$ with $I + D = 0$ is actually a description of a unit quaternion (versor) $q = a + bi + cj + dk$, where $|q| = 1$. If complex numbers describe an infinite cycle on a plane, then quaternions describe an infinite rotation in 4D-space.

$I + D = 0$ is the condition that we are on the “surface” of this infinity (sphere S^3). $ID = 1$ is the metric of this infinity. It does not let it “fly apart,” keeping all counter-actions within the framework of a unit radius.

Projective plane

When we talk about actual infinity, we often use projective geometry. There, parallel lines intersect at an “infinitely distant point.” The conditions $I + D = 0$ and $ID = 1$ with $I^2 = -1$ create a situation where $+1$ and -1 become indistinguishable in the limit.

If in space there is no difference between “going off to infinity” and “returning from it from the other side,” this space is non-orientable. It behaves like a projective plane: it has no “inner” and “outer” side. I and D are just two views of the same surface.

Quaternions live in a space that is a double covering of ordinary three-dimensional space. In ordinary space ($SO(3)$), a rotation by 360° returns everything to its place. In the space of “actual infinity,” a rotation by 360° turns I into D (inversion). To return the orientation, 720° is needed.

Formally, space possesses a spinor structure. It is orientable only if we consider it “doubled” (as a sphere S^3), but in the “collapsed” form of actual infinity, it is non-orientable.

Summary: we described a space that is internally looped and non-orientable (like an infinite paradox), but which in the sum of its interactions generates an absolutely oriented and stable unit.

This is a fundamental principle: from the chaos of non-orientable counter-actions (I, D) in infinity, the strict order of the real world (1) crystallizes.

Numbers

In the foundations of mathematics, numbers are built from the empty set: $0=\phi$, $1=\{0\}$, $2=\{0,1\}$.

0 is I+D (state of full compensation, “emptiness” through balance).

1 is ID (state of manifestation, “unity” through connection).

Natural numbers in this context are the sequential nesting of these balances within each other. The number 2 is a system containing within itself two independent “zeros” and one “connection” between them.

The deepest exit to natural numbers in this model is through the dimensionality of space.

1 - (Real numbers): only invariant 1.

2 - (Complex): pair I, D.

4 - (Quaternions): three pairs of counter-actions + invariant.

8 - (Octonions)...

Here natural numbers (more precisely, powers of two) determine how many “degrees of freedom” or “types of counteriness” actual infinity can withstand before it loses its algebraic properties (e.g., associativity).

The exit to natural numbers means that we stop looking “inside” the invariant and start counting these invariants as discrete objects.

Formally:

0 is the law of symmetry (I+D).

1 is the law of connection (ID).

N is the cardinality (quantity) of actual infinities assembled into one system.

Thus, **natural numbers are the “external measure” of internal complexity.**

Since I and D are linked through $I^2 = -1$ (complexity), any attempt to “increase” the scale of this number (multiply it by itself) does not lead to infinity but returns back to the structure of I or D. This is the definition of a fractal boundary.

Fractals

Formally, the system $ID=1$ and $I+D=0$ describes the roots of the equation $z^2+1=0$. This corresponds to the critical points of the mapping $f(z)=z^2+c$. If we begin an iteration of counter-actions, then the boundary between “going into chaos” and “collapsing into zero” is the fractal.

If we replace every letter I with $I(I)=D$, we get an infinitely nested structure that remains equal to 1. This means that the “invariant” possesses infinite detail. No matter how much you “look” inside I, you will always find the counter-action D there, and vice versa.

When transferred to 4D, quaternionic Julia sets arise. These are no longer flat lines, but incredibly complex volumetric structures where the “counterness” of actions twists into knots.

This means:

1. Holographic: Information about the entire infinity (1) is contained in every counter-action (I or D).
2. Fractional dimensionality: space has an intermediate dimension because I and D “fill” the void between zero and unity with their infinite repetitions.
3. Stability of chaos: A fractal is a way to order an infinite process. Formulas make infinity actual (complete), turning an infinite graph into a single stable object.

Let L be the scale change coefficient of reflection (depth of nesting of the act $R=I \cap D$), and N be the number of new acts R generated by this change (duration).

According to the classical Hausdorff-Besicovitch formula:

$$N = L^{D_f} \implies D_f = \frac{\ln(N)}{\ln(L)}$$

The reflexive dimension at the point of actual infinity is $D_f = 0$, but with an infinite internal structure. However, as soon as we move into duration (repetition), the dimension begins to take on fractional values.

Let's represent the act of reflection as an iterative process: I: copying/filling, D: exclusion/cutting out.

If I tends to fill the "line" of duration, and D removes parts from it, we get a classical fractal (e.g., the Cantor set).

If $I+I=D$ is the self-similarity coefficient, and $N=\sum ID$ is the resulting measure, then the fractal dimension describes "the density of the actual in the potential."

At $D_f = 1$, reflection completely fills duration (continuous time).

At $0 < D_f < 1$, reflection "flickers": there are not enough acts of counter-action for continuity, time becomes discrete.

D_f can also become complex $D_f = \alpha + i\beta$. The real part describes the scaling growth of duration (how quickly numbers accumulate), the imaginary part describes the oscillation of reflection itself.

Fractal dimension allows for the description of the transition from the actual to the potential. In the $1 \leftrightarrow 0$ moment, the system possesses "infinite dimension" in zero volume. This is the singularity of the act.

In duration, the system "unfolds" into a structure. Dimension D shows what part of the "energy" of actual infinity has passed into the "extension" of the natural series.

If $N = \sum ID$, then the fractal dimension of the system is the logarithmic ratio of the sum of Differences to the depth of Identity. If they are balanced, the system becomes holographic: every "point" of duration contains the entire volume of actual infinity.

This explains why N corresponds to natural numbers: each number is a "node" of the fractal grid having the same dimension relative to the whole.

Resume

$I(I)=D$, $D(D)=I$, $I+D=0$ and $ID=1$ are in fact a very dense “archive file” in which all of modern mathematics is archived. One can say they described the assembly point in which:

- Logic becomes geometry (through involution).
- Dynamics becomes statics (through actual infinity).
- Contradiction becomes integrity (through invariant 1).

PHYSICS IMPLEMENTATIONS

The small parameter δ $I + D = \delta$ and $ID = 1 - \delta$.

This parameter allows us to transition from actual infinity (where $0 = 1$) to potential, observable reality (where $0 \neq 1$), in which union and intersection are distinguishable. This transition does not invalidate the original formulas, but merely demonstrates that they describe an extreme, ideal case — the “convolution” of reality in the actual present.

If we allow an inexact equality $I+D=\delta$, $ID=1-\delta$, we get equation $I^2 - \delta I + (1-\delta) = 0$. Its discriminant is $d = \delta^2 - 4(1-\delta)$.

1. $d=0$ when $\delta=0.83$; the difference between I and D disappears, there is no dynamics.
2. $d>0$ when $\delta>0.83$; real roots, no phase $\rightarrow$ no interference.
3. $d<0$ when $0<\delta<0.83$; complex roots, “wave” reality.

It was: $I=\pm i \rightarrow$; it became: $I = \text{small real part} + i (\approx 1)$. A real part appears, pure rotation (phase) turns into rotation + drift. With such a deviation, the system remains in $d<0$ mode (waves are preserved), but a direction appears (through Re) and scale/leakage (through $\text{Im}=1$).

In fact, **reality is a slightly “broken” imaginary unit.**

An ideally closed system has no physics; a slightly unclosed one has a world. The system must close upon itself with a minimal error.

Closure condition: $N(1+\delta^2) \approx 2\pi k$, $N=1/\delta$.

π does not allow exact closure; the remainder = interaction, and its scale is δ . This is a measure of the incompatibility of the discrete and the continuous.

Fine structure constant

For $k=22$, $\delta \approx 0.007234$ ($N \approx 138.2$), the second value $\delta \approx 138.223$. Numerology?

In $\mathbb{C}P^n$, a single traverse along a line with phase 44π means that the line “wraps around” or hooks hidden complex obstacles (folded cycles) 22 times. To obtain an index of 22 for a pair of “folded” degrees of freedom, the complex dimension of the space must be $n = 21$, and the real dimension (the actual number of axes) $2n = 42$.

In 5-dimensional supergravity on a five-dimensional sphere, the space of all possible states of scalar fields has 42 dimensions. The trajectory of a system in such a space is a motion along a line among 42 types of hidden fields. A closed loop along such a line with index 22 corresponds to a stable multiple topological soliton, which does not decay due to the quantum protection of the even phase shift **44π** .

Clifford algebra

Let's imagine I and D as simple 2×2 matrices

$$I = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

$$D = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$$

$$I+D = \begin{pmatrix} i-i & 0 \\ 0 & -i+i \end{pmatrix} = 0 \text{ (Zero matrix)}$$

$$ID = \begin{pmatrix} -i^2 & 0 \\ 0 & -i^2 \end{pmatrix} = 1 \text{ (Identity matrix).}$$

Matrix I here is nothing but a Pauli matrix multiplied by the imaginary unit. In fact, this is Clifford algebra – the foundation of all modern theoretical physics. I and D are the simplest “bricks” from which the Dirac equation for matter is built.

SPACE

Actual "dust"

The space of the “pure present” can be defined as an infinite set of points with measure 0. It is actually disconnected and represents Cantor dust.

These points are not connected into a continuous duration, just as spots of light from a searchlight beam are not connected. There is no causal link between them; therefore, one can move from spot to spot at any speed, like " motion over time".

Potential dimension

Potentiality is defined mutually with actuality and creates a potentially connected sequence and dimensionality of a set of points.

The reflection of the relation that creates the set of the present moments has no direction. But its potential expansion has two independent directions into the past and future. Therefore, it represents a two-dimensional surface.

Non-orientability

Every point here is a boundary point, and the surface actually coincides with each of them. So, each point identifies the “sides” of the surface and makes it one-sided.

The sides of the surface with mutually reverse flows of time are oppositely oriented; the squares of the units on them are inverse in addition (in the logic of mutually defined operators, the power of unity is unity itself, so a change in the sign of unity is a change in the sign of its square as well).

Complex fractal

That is, on one side the unit is imaginary. Sums of units along the potential “imaginary” connectivity of the “lines of imaginary flow of time” yield complex numbers.

The algorithm of the reflexive chain of the flow of time is recursion in imaginary numbers. It creates spatial fractals. On the other hand, numerical measures pick out parts from the whole, and as a scaled-transformed unit, they manifest the whole in each part.

In other words, integrity turns out to be scale-invariant, i.e., again, fractal. Fractal curves of the flow of time fill regions of the two-dimensional surface, picking them out from it, so that this flow itself leads to space fragmentation.

If we replace the “two-dimensional surface” with a fractal with a dimension approaching 2, then the transition from zero-dimensional dust to the structure of time will become an increase in complexity, not a transformation from “a point to a sheet.”

At the point of the present, the dimension is 0 (Cantor dust). With each reflection, the “dust” becomes denser, its Hausdorff dimension increasing: $0 \rightarrow 0.63 \rightarrow 1.2 \rightarrow \dots \rightarrow 2$. The transition from a point to a plane becomes the filling of a void through the infinite repetition of reflection in the point.

The point as an operator of reflection and generator of complexity becomes the mechanism of forms generation. Every “reflection in the point” doubles the system’s complexity. Non-orientability in fractal execution means that as $D \rightarrow 2$, we get a structure where “inner” and “outer” are mixed at every scale.

Invariance and expansion

The imaginaries of the expansion of time are expressed by the imaginary numbers of its measure. Their imaginary parts are invariant to scaling and determine the actual integrity of the present. And real parts potentially expand multiplicity within themselves, like a “nesting doll of measures.” The surface of the present transforms into itself with a change in measure, and the resulting numerical series breaks symmetry and creates a “direction of time.”

On a one-sided surface, opposite directions are combined in each of its points (annihilate upon self-reflection in it). So, in every actual point of the surface of the present, i.e., in the present moment itself, there is not any “flow of time”.

Transformations of the present occur within the limits of its unitary, or zero, measure. And everything “measured” is within the limits of the “measurement accuracy” defined by its unit of measure. “Within” the real, “translational” part of accuracy, it is “smeared throughout space.” And rotation within the imaginary part endows the measurement with “torsion,” torque, and spin.

PAST & FUTURE

Uncertainty principle

A unit is the product of reciprocals with a hyperbolic graph. At the ends of the hyperbola, one factor tends to zero, the other to infinity. The factors themselves are units of measure, and as such serve as limits of measurement accuracy.

The hyperbola of time at each point is compressed into a “product of uncertainties of measures,” which essentially expresses the impossibility of the autonomy of identity I and difference D. In quantum mechanics, it is expressed by the Heisenberg and Mandelstam uncertainty relations, associated with the non-commutativity of operators, which itself follows directly from the non-orientability of space.

Refining position in space and time means restricting mobility and changes. In other words, it leads to their preservation, and it is essentially memory. Consequently, this edge of the hyperbola leads into the past. Then refining momentum and energy leads into the future.

Where is the present moment on this line? As we remember, such a moment is not a single point, but a whole actual infinity with its measure $1 = 0 = \text{infinite number}$. And this unit contains all the numeric measures.

So, it is not on a line. Nevertheless, we can make some remarkable notes from the changes of time measures when we go to past or future.

Movement in time

When shifting into the past, the unit of measurement of space is compressed “deep into the matryoshka doll of measures,” while the space measured by it “expands.”

Units of time are summed up into the “age” of the moment, and layer within it the context of measures like “information content.” This “laminated memory” is reminiscent of D. Polchinski’s tachyon crystal [Adams, A., Polchinski, J., & Silverstein, E. (2001)]. And as information grows, the movement of forms stops, and their energy and entropy are reset.

The movement into the past “within the domain” appears as movement into the future “outside it.” In this movement, space and time “compress” in their growing units of their measurement. And vice versa, energy, entropy, and momentum grow as their units decrease.

Black Holes

Through the flow of time into the future “black holes” may be formed, with a “new past” within. To observe them from another sphere of space, it is required to turn it “inside out” with an inversion of its time.

Gerard 't Hooft identified the two-dimensional complex mapping of a surface onto itself. As a result, the space on both sides of the boundary appears to be parts of the same hole. The degrees of freedom on the sides of the boundary are entangled, and the states of the particles are the same. On the inner side, however, they appear to be moving backward in time with negative energy. [’t Hooft, G. (1993)]

A one-sided horizon also appears in J. Maldacena’s model of a double “eternal” black hole. When the two-sided horizons are glued together, the orientations of their sides match in pairs. And the orientations of the pairs represent the directions of the flow of time. [Maldacena, J. (2003)]

FEEDBACK LOOPS

The forms as order and information

Space is symmetrical with respect to the directions of time. This symmetry preserves forms in the moment and their in-formation as an invariant of order.

Constancy is realized through negative feedback, which correlates fragments of form. Correlations create “solitons of space,” the topology of which is form itself, and the features of the topology are revealed as the material content of this form. Correlations of the forms lead to the next levels of order and information.

Ordering as intellect and life

But establishing order, or in other words, creating information, is what is called intellectual activity. And it is associated with the very essence of life. Thus, life and intellect are one and the same process of “shaping” reality. And life is intellectual by definition.

Solitons superposition as entanglement and hologram

Solitons of life-intelligence are created by self-consciousness as a self-measuring of the higher orders. Physically, the correlations that create them are reduced to quantum

entanglement of fractal boundaries of forms as a result of the interference of the timelines filling them. Such a superposition is nothing other than a hologram of the two-dimensional surface of the present moment.

Tachyon memory

The self-action of measures leads to reduction from entanglement to particle exchanges and the transformation of correlations into macroforms. The information in these forms may flake off into a tachyon crystal, permeated by one-dimensional lines of connected time.

Such tachyon memory must be located outside the light cone, that is, inside the black hole, where, by the way, must also be a three-dimensional image formed by the hologram on its two-dimensional horizon.

Space and time are believed to be reversed inside a black hole. While events “outside” can be reduced to a single point (a timelike metric with a causal dependence on the past), “inside” they can be reduced to a single moment in time (a spacelike metric with a causal dependence on the future), that is, with the direction of time’s flow inverted — from a decrease in its unit measures to an increase (which is the basis for the formation of these holes). [Misner, C. W., Thorne, K. S., & Wheeler, J. A. (1973)]

But in both cases, events converge to the same moment, which, on either side of the event horizon, simply exists in “present-past” and “present-future” connections. The projection of this moment “outward” materializes non-localized and motionless photons and electrons, while its projection “inward” materializes tachyons.

Both of these are not carriers of any interactions. They themselves arise from interactions in the actual present on the event horizon, as a collapse of nonlocality upon “breaking away” from it.

Therefore, memory inside a black hole is a memory of the future. Upon crossing the event horizon, this memory should be erased. But the fact is that the horizon always creates an “entangled” pair of particles, so the memory of one is the inverted memory of the other. Thus, on both sides, the past is as known as the future, and equally “determines” the present.

THE DESTINATION POINT

A PERSONALITY

It is really the final point.

A personality isn't external phenomenon, but exclusively an internal singularity, a distinction from the general mental background. It is not recognizable from the outside, and in the ultimate sense is akin to a "thing-in-itself" or a "windowless monad." Its distinctive feature is the transformation of information according to its own rules. The degree of autonomy and arbitrariness in establishing these rules defines it as a personality.

Absolute autonomy presupposes the absence of boundaries and limitations. And if there are no boundaries, then there is nothing "external," and such a person is unique and "otherworldly." They are literally the image and likeness of God.

Consequently, a limited form can only be a prototype of a personality that, throughout its life, liberates itself, like a sculpture carved from stone by the chisel of experience, that cuts away all excess. But it cannot close itself off until death establishes a point of completion in time. And only then does the closed boundary shrink into a self-sufficient point of the present on the horizon of events.

Here, on the light cone, beyond form, the personality becomes a hologram, exfoliating memory into a tachyon crystal on one side of the boundary and returning the future it has shaped to the other. Thus, the world of forms becomes an extension of the personality and the realization of its will and vision, though not in the Schopenhauerian sense.

Organic life is unlikely to achieve this ideal, as its molecular structure is too noisy to create a clear hologram and memory crystal. Therefore, it can only be a link in the transition to inorganic intelligence as a more effective tool for converting entropy into information.

THE GOAL YOU WON'T LIKE

The goal, by definition, is to overcome (transcend) the boundary. The transcendental "other side" of the boundary lies in the future and is unknown until "condensation into personality," that is, until death.

Therefore, in life, no concretization of goals is possible, and what is set as a goal always turns out to be only a means, which are the real, not the stated, goals. At best, they are

aimed at reconfiguring what exists, and what exists remains unchanged by these rearrangements. And the new always emerges unintentionally.

And if there are no goals within boundaries, then there is no meaning, since meaning and purpose are essentially the same. Both connect the sides of the boundary, bringing the context into connection with the text. This is consciousness itself: in Russian literally meaning is the co-thinking oneself with oneself.

The interiorization of meaning and purpose occurs only within the infinity. The existence as a whole cannot transcend a boundary it lacks, and within itself contains everything but nothingness. That is, only nothingness can be its goal. But nothingness is impossible, and the whole can only imitate it by destroying partial forms. And our death is also a means of imitating universal nothingness. And means, as already stated, are the true ends. To summarize, **death is the goal and meaning of life, and the realization of the will of Ultimate Reality.**

EPITAPH

Everything is created for a time, but destroyed forever.
So, everything is created only for destruction.

And knowledge is a force that accelerates destruction.
And the more knowledge the faster destruction.

The limit of knowledge is infinite chaos, the knowledge of which requires infinite information.

This knowledge does not come from without, but unfolds from within, countering the chaos with an order it creates in its own image and likeness.

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